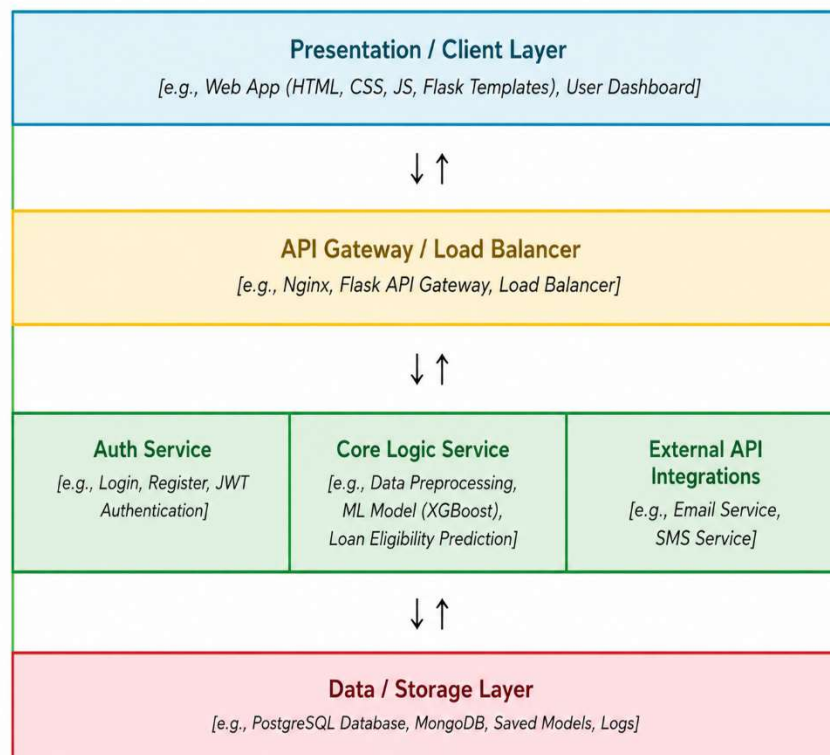


Solution Architecture

Date	29 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **Smart Lender – Loan Eligibility Prediction System** follows a layered architecture to ensure efficient, secure, and scalable loan eligibility prediction. Users interact with the system through a Flask-based web interface, where applicant details are entered. The requests are processed through the application layer, which performs data validation, preprocessing, and machine learning prediction using the trained **XGBoost** model. The business logic manages the prediction workflow and communicates with external services when required. Finally, the data layer stores the loan dataset, trained machine learning model, prediction results, and application logs. This architecture enables fast, reliable, and real-time loan approval predictions while maintaining data integrity and system performance.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation / Client Layer</i>	<i>Provides a user-friendly interface where applicants enter loan details and view the loan eligibility prediction result.</i>	<i>HTML, CSS, JavaScript, Flask Templates</i>
<i>API Gateway / Application Layer</i>	<i>Receives user requests, validates input, and forwards the data to the machine learning prediction service.</i>	<i>Flask, Python</i>
<i>Core Logic Service</i>	<i>Performs data preprocessing, feature encoding, and executes the trained XGBoost model to predict loan eligibility.</i>	<i>Python, Scikit-learn, XGBoost, Pandas, NumPy</i>
<i>External API Integrations</i>	<i>Supports future integration with banking systems, credit verification services, or notification services if required.</i>	<i>REST APIs (Optional)</i>
<i>Data / Storage Layer</i>	<i>Stores the loan dataset, trained machine learning model, prediction results, and application logs.</i>	<i>CSV Dataset, Pickle (.pkl), IBM Cloud Storage</i>